

NETWORK TROUBLESHOOTING & PACKET ANALYSIS

Wireshark hands-on lab | Technical portfolio case study

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Date: October 8, 2026

Tools: Wireshark, Windows Command Prompt, nslookup, ping, curl

Executive summary

This independent lab investigated DNS resolution, ICMP reachability, TCP connection establishment, and TLS-protected traffic using Wireshark packet captures. Evidence discussed below comes from screenshots supplied during the guided exercise. Five associated PCAPNG files were also provided; their contents have not been independently decoded in this report. Findings are limited to the observed exchanges, not a comprehensive assessment of network health.

Lab objectives

Identify request/response traffic; distinguish IPv4 and IPv6 DNS lookups; correlate ICMP echo requests and replies; verify a TCP SYN / SYN-ACK / ACK sequence; and identify TLS handshake and application-data records.

1. DNS resolution

The DNS screenshot showed A and AAAA queries for example.com and their corresponding responses. The expanded DNS response for packet 491857 showed transaction ID 0x0008, status No error, one question, two answer resource records, and a displayed response time of 154.99 ms. This establishes that the observed DNS transaction returned successfully; it does not establish the performance of DNS over time.

Evidence	Observed result
A query / response	Packets 487844 / 491857
AAAA query / response	Packets 491939 / 497793
Response metadata	ID 0x0008; No error; 2 answer RRs; 154.99 ms

2. ICMP reachability

The ICMP screenshot showed four echo requests from 10.0.0.205 to 1.1.1.1 and four matching echo replies. This demonstrates successful reachability during the captured test, with no missing replies among those four exchanges. The reply TTL was 58. A four-packet sample cannot establish sustained zero packet loss.

Metric	Observed
Echo requests / replies	4 / 4
Observed reply rate	100% within sample
Approximate round-trip times	23.6, 18.9, 17.9, 25.1 ms
Approximate average	21.4 ms

3. TCP three-way handshake

The screenshot captured three consecutive TCP packets to port 80: packet 440 [SYN], packet 441 [SYN, ACK], and packet 442 [ACK]. These three messages verify completion of the connection-establishment handshake in that exchange. The time from first SYN to final ACK was approximately 0.9 ms based on the displayed timestamps.

Packet	Flags	Role
440	SYN	Client initiates connection
441	SYN, ACK	Server responds
442	ACK	Client completes handshake

4. TLS handshake and encrypted records

The TLS screenshot identified packet 273 as Client Hello (SNI=example.com) and packet 275 as Server Hello / Change Cipher Spec. Another screenshot using the display filter `tls.record.content_type == 23` showed multiple TLS Application Data records. A selected frame displayed TCP source port 56729 and destination port 443. These observations establish TLS negotiation and the presence of encrypted records in the capture, but do not alone verify successful delivery of the particular HTTPS request.

Note: Wireshark may label TLS 1.3 encrypted records as TLSv1.2 due to record-layer compatibility fields. The visible payload bytes should not be interpreted as readable application content.

5. Troubleshooting relevance

DNS: Check whether a hostname resolves and whether the resolver returns errors.

ICMP: Test IP reachability independently of name resolution.

TCP: Identify whether a connection request receives a SYN-ACK and completes with ACK.

TLS: Confirm whether a client initiates negotiation and inspect protocol metadata without assuming encrypted payload visibility.

Evidence inventory

Capture file	Size
01_Initial_DNS_small_Capture.pcapng	19,708 bytes
02_DNS_small_Lookup.pcapng	1,024 bytes
03_ICMP_Ping.pcapng	109,048 bytes
04_TCP_Three_Way_Handshake.png.pcapng	188,884 bytes
06_TLS_Encrypted_Application_Data.png.pcapng	163,620 bytes

The conversation included screenshots of DNS packet lists and response details, ICMP packets, the TCP three-packet sequence, and TLS handshakes and application-data records. The original image attachments were not all available as separate image files for PDF embedding. Accordingly, this report summarizes the visible evidence rather than misrepresenting recreated graphics as original screenshots.

Portfolio / resume description

Wireshark Network Troubleshooting & Protocol Analysis Lab — Captured and inspected DNS, ICMP, TCP, and TLS network traffic; correlated DNS requests with responses, verified ICMP echo exchanges, identified a complete TCP three-way handshake, and analyzed TLS handshake metadata and encrypted record types. Documented packet-level findings and the limitations of each diagnostic method.

Privacy and next steps

Retain raw PCAPNG files privately. Before publishing screenshots, redact local and public IP addresses, MAC addresses, hostnames, session identifiers, and other potentially identifying details. For a public GitHub portfolio, publish this report or a sanitized version rather than unreviewed raw captures.